

María Touceda-Suárez, PhD

Postdoctoral Researcher

Bioinformatics Research Center (BRC) – North Carolina State University (NCSU)

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Postdoctoral researcher driven by open science, reproducibility, and FAIR data principles – believes good science should be usable and verifiable by others, not just publishable. Works across viral, bacterial, and plasmid metagenomics in environmental microbial communities, driven to find solutions and adaptations to climate change that improve outcomes for people and the planet. As Nextflow Ambassador (2026), leads workshops and training materials that take graduate students and professionals from zero to running their own pipelines.

EMPLOYMENT

Postdoctoral Researcher

September 2025 – Present

Bioinformatics Research Center (BRC) - North Carolina State University (NCSU)

Raleigh, NC

- Developing a reproducible Nextflow pipeline to systematically benchmark ML-based viral inference tools (e.g., geNomad, VirSorter2, DeepVirFinder) against realistic metagenomic challenges, scoring outputs against simulated communities with known ground truth to quantify precision/recall and identify systematic failure modes
- Designing and building lab-wide computational infrastructure, including Nextflow pipeline logic and a structured data lake, to support scalable multi-omics workflows
- Leading viral metagenomic analysis of European wastewater samples — sequence assembly, viral identification, and annotation — to characterize wastewater viral community composition; extending analysis to plasmid detection and characterization
- Leading viral and plasmid analysis of bee gut microbiome in a passage experiment to study the effect of drift on microbial communities
- Develop and teach bioinformatics curricula (metagenomics, RNA-Seq) for NC State graduate students and veterinary science audiences at University College Dublin, translating complex genomic workflows into accessible technical instruction
- Serve as go-to technical resource for faculty members and students on metagenomics pipeline design and troubleshooting, providing hands-on consultation on tool selection and statistical approach
- Developed a full bioinformatics training curriculum from the ground up — HPC usage, Unix/Bash scripting, metagenomics processing, and Nextflow
- Deploy nf-core pipelines as self-service Open OnDemand apps on institutional HPC via nf2ood

Graduate Assistant

January 2020 – May 2025

University of Arizona (UofA)

Tucson, AZ

- Led viral and plasmid metagenomic analysis of soil samples on university HPC infrastructure (Barberán Lab), including full pipeline execution and troubleshooting of sequence assembly/annotation errors
- Designed and delivered bioinformatics pipelines for soil and environmental metagenomics
- Led field soil sampling and DNA extraction for soil viral and microbial community studies
- Instructor of record and TA for undergraduate data analysis and environmental science courses (100–200 level), consistently translating quantitative/genomic concepts for non-specialist students

Junior Programmer

February 2019 – May 2019

Sopra Steria

Madrid, Spain

- Back-end application programming

Graduate Assistant

Technical University of Denmark (DTU)

June 2017 – March 2018

Copenhagen, Denmark

- Curated microbial tool metadata for Elixir bio.tools; supported nitrification-anammox wastewater bioreactor research at DTU Environment

EDUCATION

PhD in Environmental Science

University of Arizona (UofA)

January 2020 – May 2025

Tucson, AZ

- PhD Dissertation: “Harnessing metagenomics to unveil the underrepresented fraction of the soil microbiome”. Supervisor: Dr. Albert Barberán
- Student Representative CALS GPSC (2021-2023)

MSc in Bioinformatics

University of Copenhagen (UCPH)

November 2018

Copenhagen, Denmark

- Master Thesis – DTU Environment. “Network construction and analysis from microbial community meta-omics data”. Supervisor: Dr. Barth F. Smets. Co-authored resulting publication (Environmental Microbiome, 2022)

BSc in Human Biology

Pompeu Fabra University (UPF)

July 2016

Barcelona, Spain

- Bachelor Thesis – Systems Biology CRG. Supervisor: Dr. Manuel Irimia
- Student Representative

SKILLS

Computational: Python, R, Bash; HPC pipeline parallelization (SLURM, LSF, PBS); Nextflow, Snakemake; GNU Parallel, Chapel; Git/GitHub (including GitHub Actions); Basic app/web development (Java, JavaScript, Shiny)

Data Science: Supervised/unsupervised ML, Data visualization, Data mining, Data curation, FAIR data principles

Bioinformatics: NGS data cleaning, Assembly, and annotation; Metagenomics, metatranscriptomics, and metabolomics analysis; Functional and taxonomic annotation (bacteria, fungi); Amplicon sequencing analysis; Viral and mobile genetic element inference from metagenomes; Microbial ecology and network analysis

Environmental Data Science: Soil/water physicochemical analysis, mass spectrometry data curation, survey/human-subjects studies, basic GIS

Software & Tools:

- Design & figures: Adobe Illustrator, Canva, BioRender
- Authoring & documentation: Quarto, R Markdown, LaTeX, PowerPoint
- Development environments: RStudio, Jupyter, VS Code
- Collaboration & project management: Slack, Notion, Agile development practices
- AI-assisted development: Claude and Claude Code, GitHub Copilot

Laboratory & Field: Soil sampling, DNA extraction, PCR, microbial culture

Languages: Spanish (native), Galician (native), English (fluent), Catalan (conversational), Portuguese (conversational)

TEACHING AND MENTORING

Mentoring of Graduate, Undergraduate, and high School Students

North Carolina State University, USA

- Sadaf Raoufi (PhD student) 2025-present
- Naomi Kolodisner (PhD student) 2025-present
- Elizabeth Nitz (PhD student) 2025-present
- Li Ye (PhD student) 2025-present

University of Arizona, USA

- Carson Scholars mentor 2023- 2024
- Julia Rudolph (Accelerated masters project) 2020-2023
- Aaron Tews (Accelerated masters project) 2020-2023
- Gabriela Iñigo (Masters project) 2021- 2025
- Adalee Martínez (Masters project) 2021- 2023
- Maddie Morris (Bachelors honors project) 2020-2022

Teaching

- Invited lecturer: GN 427 – Introductory Bioinformatics FALL 2026
- Instructor of record: ENVS 275 – Data Analysis for Life and Environmental Sciences. FALL 2023
- Teaching assistant: ENVS 275 – Data Analysis for Life and Environmental Sciences. FALL 2022
- Teaching assistant: ENVS 107 – Introduction to Environmental Science. SPRING 2022
- Teaching assistant: ENVS 275 – Data Analysis for Life and Environmental Sciences. FALL 2021

Workshops

- BRC's Nextflow Workshop, North Carolina State University, Raleigh, NC October 2026
 - Omics with R, North Carolina State University, Raleigh, NC August 2026
 - Metagenomics for Veterinary Science, University College Dublin, Dublin, Ireland June 2026
 - Introduction to Hazel HPC - Week Intensive, North Carolina State University, Raleigh, NC May 2026
 - Introduction to Hazel HPC, North Carolina State University, Raleigh, NC Septemeber 2025-May 2026
 - Soil microorganisms and Their Role in Soil Health, Mission Garden, Tucson, AZ October 2024
 - Complex Systems Science, University of Arizona, Tucson, AZ January-February 2023
- Designed and led a two-day workshop on complexity science and network analysis for graduate students in environmental and ecological sciences.

COMMUNITY OUTREACH

Volunteer

Living Streets Alliance, Tucson, USA — October 2022 – May 2025

- Assisted with traffic control for the Cyclovia open-streets event (2022)
- Led the data gathering and data analysis/visualization teams for Cyclovia (2023–2025)

Department of Environmental Science, University of Arizona, Tucson, USA

2021-2022

- Supported invasive grass eradication campaigns at Saguaro National Park
- Participated in cleanup campaigns at the Arroyo Chico Wash

Mission Garden Community Historical Garden, Tucson, USA

2020-2021

- Served as event docent, educating visitors on agricultural history and local flora
- Assisted with garden operations — harvesting, building, and event organizing

The Community Food Bank, Tucson, USA

December 2020

- Supported food distribution logistics for community outreach

Coordination

May 2021

ResBazz (Research Bazaar) 2021, University of Arizona, USA

Tucson, USA

Supported the ResBazz event support team and co-led data science literacy workshops

TALKS AND PRESENTATIONS

Poster presentations

- October 2026 – ASMBIG – Bioinformatics, Genomics, and Big Data Conference, Washington DC, USA
Title: “Bacteriophage survival in the bee microbial gut along a passage experiment”
- March 2026 – Envisioning Urban Futures, North Carolina State University, NC, USA
Title: “Foundations for Change: Urban Soil Bacterial and Viral Exploration as a Platform for Education and Women's Computational Training”
- June 2024 – International Soil Virus Conference, Lawrence-Livermore National Laboratory, Livermore, CA, USA
Title: “Soil viral response to drought disturbance and ecosystem recovery in an artificial rainforest”
- March 2023 – ENVISION, University of Arizona, Tucson, USA
Title: “Urbanization causes major changes in the functional profile of arid soil metagenomes”
- August 2022 – ISME’18, Lausanne, Switzerland
Title: “Urbanization causes major changes in the functional profile of arid soil metagenomes”
- July 2018 – ISME’17, Leipzig, Germany
Title: “Network construction and analysis from microbial community meta-omics data”

Oral presentations

- March 2024 – ENVISION, University of Arizona, Tucson, USA
Title: “Differences in the genetic potential of soil microbial communities of urban greenspaces vs. natural arid soils”. Honorable Mention
- March 2024 – Mission Garden lecture series, Tucson, USA
Title: “Soil health presentation: Microorganisms in Mission Garden soil”
- March 2022 – ENVISION, University of Arizona, Tucson, USA
Title: “Effects of historical agricultural crops on the soil microbiome of a living agricultural museum in the Sonoran Desert”
- April 2021 – Women in Data Science, Tucson, USA
Title: “Effect of land use in arid soil microbial communities”
Winner of the data BLITZ competition
- March 2021 – ENVISION, University of Arizona, Tucson, USA
Title: “Soil microorganisms in Tucson agricultural history”

RECENT GRANTS AND AWARDS

Travel Grant

2024

Department of Environmental Science, University of Arizona

- Awarded \$2,500 to fund travel to the International Soil Virus Conference, at the Lawrence-Livermore National Laboratory, CA, USA

Summer Research Fellowship

2023

Department of Environmental Science, University of Arizona

- Awarded \$7,000 to fund summer research stay at the University of Helsinki, Finland.
- Awarded the funds in recognition for achievements and service to the department.

Impact Leader Fellowship

2023

Bayer Crop Sciences, Porterbay Insight and the CALS Career Center, University of Arizona, USA

- Professional development training fellowship awarded to ~20 graduate students a year.
- Provides students with skills employers increasingly desire in prospective candidates.
- The fellowship included monthly workshops and networking with industry mentors.

Graduate and Professional Student Council (GPSC) Travel Grant

2022

GPSC, University of Arizona

- Awarded \$1,500 to fund summer conference travel

Complex Systems Summer School

2022

Santa Fe Institute, USA

- Awarded to 60 applicants internationally
- One-month intensive summer school that included talks and seminars from multidisciplinary scientists, workshops, and interdisciplinary project development

Carson Scholars Fellowship

2022-2023

Arizona Institute for Resilient Environments & Societies (AIRES), University of Arizona, USA

- Awarded to 10 graduate University of Arizona students a year
- Focused on training scholars in science communication
- The fellowship included monthly science communication workshops, a writing retreat, and a \$5,000 stipend
- Review panel member for the 2023-2024 cohort
- Guest at the Planet Forward Summit 2023 – Washington D.C.

Roots For Resilience Fellowship

2021

Institute for Resilience (AIR), University of Arizona, USA

- Awarded to 10 graduate students a year
- Focused on increasing Open Science literacy across the University of Arizona
- The fellowship included .25 FTE and registration to the CyVerse Open Science Workshop

RELEVANT PUBLICATIONS

- **Touceda-Suárez, M.,** Ponsero, A.J., Barberán, A. Urban greenspaces favor heavy metal resistance and competitive traits in plasmids from arid soils. *Microbiology* 172, 001705 (2026).
<https://doi.org/10.1099/mic.0.001705>

- Morris, M.L., Eastburn, D.J., Roche, L.M., Davy J., Doran, M., Karle, B., Lile, D., Schohr, T., Snell, L., Macon, D., Woodmansee, G., **Touceda-Suárez, M.**, Barberán, A. Effects of grazing exclusion and pasture management on soil microbial communities in Californian irrigated pastures. *Applied Soil Ecology* 215, 106419 (2025). <https://doi.org/10.1016/j.apsoil.2025.106419>
- Iñigo-Gámiz, G., **Touceda-Suárez, M.**, Bustamante, E. et al. Above and belowground effects of buffelgrass invasion in the Sonoran Desert. *Biol Invasions* 27, 195 (2025). <https://doi.org/10.1007/s10530-025-03650-0>
- **Touceda-Suárez, M.**, Perry, M. A., Frizzo, R., Lotz-McMillen, J. H., Gilmore, R. A., Bennett, S. M., Basso, J. T. R., Donovan, W., Fudyma, J. D., Geonczy, S. E., Gittrich, M., Gogul, G., Hazard, C., Hillary, L. S., Jameson, E., Jiraska, L., Johnson, S. S., Kosmopoulos, J. C., Lelewi, I., ... Trubl, G. Meeting report: International soil virus conference 2024. *Virus Research* 354, 199544 (2025). <https://doi.org/10.1016/j.virusres.2025.199544>
- **Touceda-Suárez, M.**, Ponsero, A.J., Hernandez, L.K., Gil-Loaiza, J., Fudyma J.D., Barberán, A., Meredith, L.K. Soil viral response to drought disturbance and ecosystem recovery in an artificial rainforest. (Submitted to *Soil Biology and Biochemistry*)
- **Touceda-Suárez, M.**, Ponsero, A.J., Barberán, A. Differences in the genomic potential of soil bacterial and phage communities between urban greenspaces and natural arid soils. *Applied Environmental Microbiology* 91:e02124-24 (2025). <https://doi.org/10.1128/aem.02124-24>
- Touceda-Suárez, A., **Touceda-Suárez, M.**, Arbolea, J.C., Sörensen, P.M. Harnessing *Bacillus amyloliquefaciens* for Amazake Production: Comparison with *Aspergillus oryzae* Amazake for Metabolomic Characteristics, Microbial Diversity, and Sensory Profile. *Foods* 13, 13 (2024). <https://doi.org/10.3390/foods13132012>
- **Touceda-Suárez, M.**, Cowan, D., Rockey, E., Barberán, A. Effects of agricultural crops on the soil microbiome of a living agricultural museum in the Sonoran Desert. *Agronomy Journal* (2023). <https://doi.org/10.1002/agj2.21456>
- Palomo, A., Azevedo, D., **Touceda-Suárez, M.**, Domingo-Félez, C., Mutlu, A.G., Dechesne, A., Wang, Y., Zhang, T., Smets, B.S. Efficient management of the nitrification-anammox microbiome through intermittent aeration: absence of the NOB guild and expansion and diversity of the NOx reducing guild suggests a highly reticulated nitrogen cycle. *Environmental Microbiome* 17, 39 (2022). <https://doi.org/10.1186/s40793-022-00432-2>
- **Touceda-Suárez, M.**, Kita, E.M., Acemel, R.D., Firbas, P.N., Magri, M.S., Naranjo, S., Tena, J.J., Gómez-Skarmeta, J.L., Maeso, I., Irimia, M. Ancient genomic regulatory blocks are a source for regulatory gene deserts in vertebrates after whole-genome duplications, *Molecular Biology and Evolution* 10, 37 (2020). <https://doi.org/10.1093/molbev/msaa123>